

Exercise 1

Part A - Intro to Logic

1 Propositional Logic

Exercise 1 (Sheffer Stroke). The operator $\uparrow$, also known as Sheffer stroke or NAND gate, has the following truth table:

$\uparrow$	0	1
0	1	1
1	1	0

Show that any propositional formula with language $\{\wedge, \vee, \rightarrow, \neg, 0, 1\}$ can be equivalently expressed using only the language $\{\uparrow\}$

Exercise 2 (Decision Procedure for SAT). Recall the “Decision and Simplification” proof system (Infer_D):

$$\frac{F \quad G}{F[x := 0] \vee G[x := 1]} \text{CA} \qquad \frac{F}{F'} \text{SIMP with } F \rightsquigarrow F'$$

where $\rightsquigarrow$ is the result of applying a finite sequence of any of the following simplifications anywhere in the formula:

$$\begin{array}{llll} 0 \wedge F \rightsquigarrow 0 & F \wedge 0 \rightsquigarrow 0 & 1 \wedge F \rightsquigarrow F & F \wedge 1 \rightsquigarrow F \\ 0 \vee F \rightsquigarrow F & F \vee 0 \rightsquigarrow F & 1 \vee F \rightsquigarrow 1 & F \vee 1 \rightsquigarrow 1 \\ \neg 0 \rightsquigarrow 1 & \neg 1 \rightsquigarrow 0 & & \end{array}$$

For each of the following formulas, express it using only the connectors $\neg$, $\wedge$ and $\vee$. Then prove they are tautologies by assuming their inverse and deriving 0 using Infer_D .

1. $((a \rightarrow b) \rightarrow a) \rightarrow a$
2. $\neg(a \wedge b) \rightarrow (\neg a \vee \neg b)$
3. $((\neg a \rightarrow b) \wedge (a \rightarrow b)) \rightarrow b$

Exercise 3 (if-then-else).

1. Consider the two following small programs:

```
1  def f1(a:Boolean, b:Boolean, c:Boolean): Boolean = {
2    if a || b then
3      if b && !a then
4        b && c
5      else
6        !b && c
7    else
8      c
9  }
```

```
1  def f2(a:Boolean, b:Boolean, c:Boolean): Boolean = {
2    if c then
3      if a then
4        !b
5      else true
6    else false
7  } ensuring (_ == f1(a, b, c))
```

Are they equivalent on all inputs? Show that they are or aren't by expressing those programs as propositional formulas. Note that the ensuring clause is automatically proved or disproved by Stainless!

2. Can any formula be expressed with only **if a then b else c**, where a, b and c are recursively constructed the same way? What if a is restricted to be a variable?

Exercise 4 (Propositional Tautologies). For each of the following propositional logic formulas determine whether it is valid or not. If it is valid, prove it (using any acceptable mathematical argument of your choice), otherwise give a counterexample. (From *The Calculus of Computation* by A.R. Bradley and Z. Manna.)

1. $(P \wedge Q) \rightarrow P \rightarrow Q$
2. $(P \rightarrow Q) \vee (P \wedge \neg Q)$
3. $(P \rightarrow Q \rightarrow R) \rightarrow P \rightarrow R$
4. $(P \rightarrow (Q \vee R)) \rightarrow P \rightarrow R$
5. $\neg(P \vee Q) \rightarrow R \rightarrow \neg R \rightarrow Q$
6. $(P \rightarrow Q) \rightarrow P \rightarrow Q$

7. $((P \rightarrow Q) \rightarrow P) \rightarrow P$
8. $((P \rightarrow Q) \rightarrow P) \rightarrow Q$
9. $(\neg Q \rightarrow \neg P) \rightarrow P \rightarrow Q$
10. $(\neg R \rightarrow \neg Q \rightarrow \neg P) \rightarrow P \rightarrow Q \rightarrow R$
11. $(P \vee Q) \rightarrow (P \vee \neg Q) \rightarrow (\neg P \vee \neg Q) \rightarrow P$
12. $(P \vee Q) \rightarrow (P \vee \neg Q) \rightarrow (\neg P \vee \neg Q) \rightarrow Q$
13. $\neg(P \wedge Q) \rightarrow P \rightarrow \neg Q$

2 Transition Systems and Invariants

Notation: $M = (S, I, r, A)$ is a transition system where S are the states, $I \subseteq S$ the set of initial states, $r \subseteq S \times A \times S$ and A is the input alphabet.

Exercise 5 (Special Invariants). Prove the following:

1. S is an inductive invariant. It is the largest among all invariants.
2. $Reach(M)$ is an inductive invariant.
3. $Reach(M)$ is the smallest of all possible invariants.

Exercise 6 (Closure of Invariants). A set is closed under an operation if applying the operation to elements of the set gives the result in the set. For example, the set of even natural numbers is closed under addition, whereas the set of odd natural numbers is not closed under addition.

Is the set of all invariants of M closed under the following operations:

1. union
2. intersection
3. complement with respect to S
4. operation $f : 2^S \rightarrow 2^S$ defined by $f(X) = Reach(M) \cup (S \setminus X)$

Answer the same questions about all *inductive invariants* of M .

3 Relations

We consider relations $r, s, t, r_1, r_2, r'_1, \dots \subseteq A \times A$ (where A is just arbitrary set, nothing to do with the input signals of the transition systems). Here $X \subseteq A$. Define:

$$\begin{aligned}\Delta_X &= \{(x, x) \mid x \in X\} \\ r^{-1} &= \{(x, y) \mid (y, x) \in r\} \\ X \bullet r &= r[X] \\ \text{ran}(r) &= \{y \mid \exists x. (x, y) \in r\}\end{aligned}$$

Exercise 7 (Relation Identities). Prove the following or give a counterexample.

1. $(X \bullet r_1) \bullet r_2 = X \bullet (r_1 \circ r_2)$
2. $(r \cup s) \circ t = (r \circ t) \cup (s \circ t)$
3. $(r \cap s) \circ t = (r \circ t) \cap (s \circ t)$
4. $(r_1 \circ r_2)^{-1} = (r_2^{-1} \circ r_1^{-1})$
5. $X \bullet r = \text{ran}(\Delta_X \circ r)$
6. If $r_1 \subseteq r'_1$ then $r_1 \circ r_2 \subseteq r'_1 \circ r_2$ and $r_2 \circ r_1 \subseteq r_2 \circ r'_1$.
7. If $r_1 \subseteq r'_1$ then $r_1 \cup r_2 \subseteq r'_1 \cup r_2$ and $r_2 \cup r_1 \subseteq r_2 \cup r'_1$.

Exercise 8 (Transitive relations). Given a relation $r \subseteq A \times A$, prove that r is transitive if and only if $r \circ r \subseteq r$.

Exercise 9 (Symmetric relations). Recall that a relation $r \subseteq A \times A$ is symmetric if $\forall x, y \in A. (x, y) \in r \Rightarrow (y, x) \in r$.

Now let r be an arbitrary relation. Prove that $r^{-1} \circ (r \cup r^{-1})^* \circ r$ is symmetric.

Exercise 10 (Transitive closure). Recall that we define the powers of a relation $r \subseteq A \times A$ as follows:

$$r^0 = \Delta_A, \quad r^1 = r, \quad \text{and} \quad r^{n+1} = r^n \circ r$$

We showed that the *reflexive and transitive closure* $r^* = \bigcup_{n \geq 0} r^n$ is the smallest reflexive and transitive relation on A containing r . Show that for any relation r on a set A , $(r \cup r^{-1})^*$ is the least equivalence relation containing r . Precisely, show that

- (i) $(r \cup r^{-1})^*$ is an equivalence relation, and
- (ii) if s is an equivalence relation containing r , then $(r \cup r^{-1})^* \subseteq s$.

4 Finite State Machines with Boolean Variables

Exercise 11 (Finite State Machines with Boolean Variables). We consider in this exercise finite-state machines enriched with boolean variables (FSM for short). Formally, an FSM is a pair (V, Q, δ) where:

- V is a finite set of (boolean) variable names,
- Q is a finite set of states,
- For every pair of states p and q , $\delta(p, q)$ is a propositional formula containing variables from V and V' , where V' is a copy of V with primed variable names.

A configuration of the FSM is determined by a pair (q, m) where

- q is a state in Q , called the *control state*,
- $m : V \rightarrow \{\top, \perp\}$ is a mapping from variable names to true or false.

The FSM can move from a configuration (p, m) to (q, m') if the formula $\delta(p, q)$ is true when we interpret every variable from V using the mapping m , and every variable in V' using m' . We define the relation $r \subseteq (Q \times V \rightarrow \{\top, \perp\}) \times (Q \times V \rightarrow \{\top, \perp\})$ to contain all such pairs of configurations (p, m) and (q, m') .

Consider now the FSM in Figure 1, inspired from [Peterson's algorithm](#) (we recommend reading this page before the exercise). We made use of two conventions when drawing the figure:

- When there is no transition drawn from a state p to a state q , we mean that $\delta(p, q) = \perp$ in the FSM.
- When a primed variable X' does not appear in a transition, we leave the variable unchanged, meaning that there is an implicit conjunct $X' \leftrightarrow X$ in the transition.

For example, the transition from $(0, 0)$ to $(0, 1)$ can be taken regardless of the initial mapping of boolean variables. It changes variable A to $\top$, and leaves the variables B and C unchanged. The transition from $(2, 0)$ to $(3, 0)$ can only be taken when the mapping of boolean variables respects $\neg A \vee C$, and leaves all variables unchanged.

Does there exist mappings m and m' such that the FSM can move (using multiple steps) from configuration $((0, 0), m)$ to $((3, 3), m')$, i.e. such that $((0, 0), m), ((3, 3), m') \in r^*$?

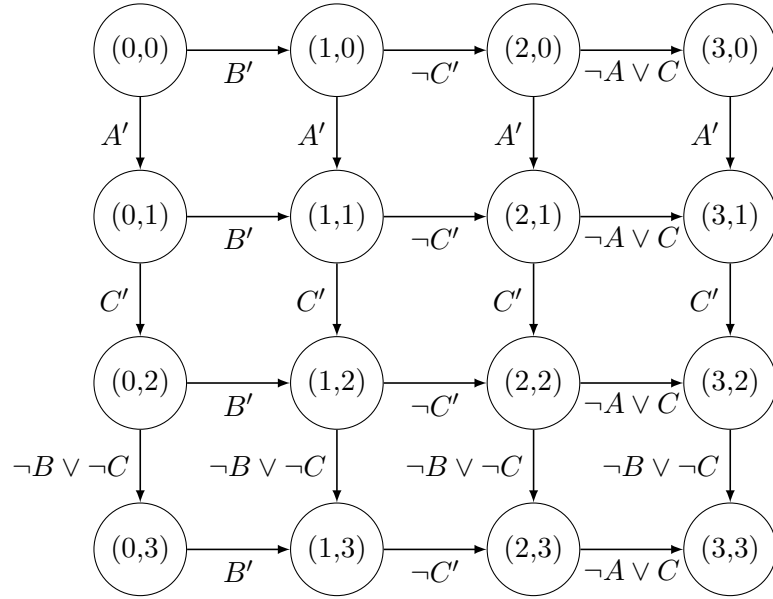


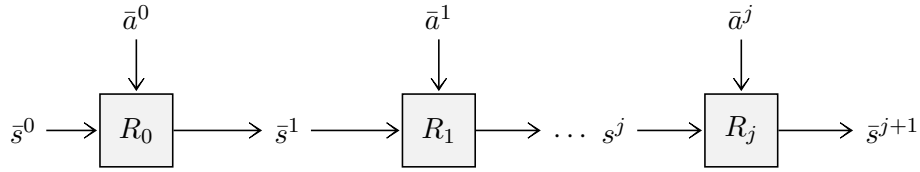
Figure 1: A finite-state machine with boolean variables $\{A, B, C\}$.

Part B - More Logic

Exercise 12 (Traces). Recall formulas T_j defined in the lecture that are satisfiable if and only if there exists a trace of length j starting from initial state whose final state satisfies the error formula E , where $FV(E) \subseteq \{s_1, \dots, s_n\}$.

$\bar{s}^i$ denotes state variables in step i .

$\bar{a}^i$ denotes inputs in step i .



$$T_j \equiv \text{Init}[\bar{s} := \bar{s}^0] \wedge \left(\bigwedge_{i=0}^{j-1} R_i \right) \wedge E[\bar{s} := \bar{s}^j]$$

where R_i is our transition formula, with variables renamed:

$$R_i \equiv R[(\bar{s}, \bar{a}, \bar{x}, \bar{s}') := (\bar{s}^i, \bar{a}^i, \bar{x}^i, \bar{s}^{i+1})]$$

1. In this system (which has n Boolean variables), give a simple upper bound on the longest trace from the initial state that does not contain any repeated states.

Let the set of error states be $S_E = \{\bar{v} \mid [\bar{s} \rightarrow \bar{v}] \models E\}$.

2. Use T_j and similar formulas to find quantified Boolean formulas F (QBF, with quantification only over propositional variables) whose free variables are $s_1, \dots, s_n$ such that the set of states characterized by F :

$$\{\bar{v} \mid [\bar{s} \mapsto \bar{v}] \models F\}$$

is exactly the set of:

- (a) states that can be reached in exactly k steps, $(\bar{r}^k)[\text{Init}]$
 - (b) states that can be reached in at most k steps, $(\bigcup_{i=0}^k \bar{r}^i)[\text{Init}]$
 - (c) all reachable states $(\bigcup_{i=0}^{\infty} \bar{r}^i)[\text{Init}]$
 - (d) states that can reach an error state in exactly k steps, $(\bar{r}^{-1})^k[S_E]$
 - (e) states that can reach an error state in at most k steps, $(\bigcup_{i=0}^k (\bar{r}^{-1})^i)[S_E]$
 - (f) all stuck states in the system, $\{\bar{v} \mid \neg \exists \bar{v}'. (\bar{v}, \bar{v}') \in \bar{r}\}$
3. Write quantified Boolean formulas *without free variables* expressing that:

- (a) no error state is reachable from the initial state
- (b) no stuck state is reachable from the initial state
- (c) every state in the system is reachable in k or fewer steps
- (d) there exists an infinite trace of the system (for some behavior of the environment given by the values of $\bar{a}$ in every step).

Exercise 13 (Tseytin Transformation). The Tseytin transformation of a propositional formula ϕ is a formula in Conjunctive Normal Form (CNF) produced the following way: recursively for each subformula ϕ_k of ϕ of the form $x_i \circ x_j$, create a new variable y_k and a new formula $y_k \leftrightarrow x_i \circ x_j$ asserting the equivalence of the two (for $\circ \in \{\vee, \wedge, \rightarrow\}$ and similarly for negation). Replace ϕ_k by y_k in ϕ and continue until ϕ is reduced to a single variable. Take the conjunction of that variable with all created assignments. For example,

$$\neg(a \wedge b) \rightarrow c$$

becomes

$$(y_1 \leftrightarrow (a \wedge b)) \wedge (y_2 \leftrightarrow \neg y_1) \wedge (y_3 \leftrightarrow y_2 \rightarrow c) \wedge y_3$$

Any expression of the type $y_k \leftrightarrow x_i \circ x_j$ can then also be equivalently expressed in CNF.

A formula can always be transformed into an equivalent one in conjunctive normal form without adding new variables, but doing so may require exponential increase in size. The Tseytin transform is guaranteed to increase the size of any formula by only a constant factor.

1. Compute the Tseytin transformation of $\neg(((a \vee \neg b) \wedge \neg(b \rightarrow c)) \rightarrow a)$.
2. Prove that on all inputs, a formula and its Tseytin transform are always equisatisfiable (one is satisfiable if and only if the other is), but not necessarily equivalent.
3. Prove that if Tseytin's transform of $\neg F$ is unsatisfiable, then F is a valid formula.
4. Given that the size (number of subformulas) of a formula is n , compute the asymptotic size of its Tseytin transformation, in Big O notation.

Exercise 14 (SAT solving). Prove the validity of the formula

$$((a \vee \neg b) \wedge \neg(b \rightarrow c)) \rightarrow a$$

by first using the Tseytin's transformation you have computed in the first question of Exercise 2, and then following by a resolution proof on clauses to obtain an empty clause.

Exercise 15 (First-order formulas). For each of the following pairs (A, B) of first-order formulas, decide whether:

- (a) $A \models B$, i.e. if B holds in any structure that satisfies A , or
- (b) $B \models A$.

Give either a proof if true or a counterexample if false for each case.

- | | |
|---|---|
| 1. $A : \forall x.(P(x) \vee Q(x))$
$B : (\forall x.P(x)) \vee (\forall x.Q(x))$ | 4. $A : \exists x.(P(x) \wedge Q(x))$
$B : (\exists x.P(x)) \wedge (\exists x.Q(x))$ |
| 2. $A : \exists x.(P(x) \vee Q(x))$
$B : (\exists x.P(x)) \vee (\exists x.Q(x))$ | 5. $A : \forall x.(P(x) \rightarrow Q(x))$
$B : (\forall x.P(x)) \rightarrow (\forall x.Q(x))$ |
| 3. $A : \forall x.(P(x) \wedge Q(x))$
$B : (\forall x.P(x)) \wedge (\forall x.Q(x))$ | 6. $A : \exists x.(P(x) \rightarrow Q(x))$
$B : (\exists x.P(x)) \rightarrow (\exists x.Q(x))$ |

Exercise 16 (First-order structures). Consider a first-order language (signature) $\mathcal{L} = \{L, B, T\}$ with one relation symbol, L , of arity two, and constants B and T . Consider the following formulas in the language $\mathcal{L}$:

$$\begin{aligned} \text{Min} : & \quad \forall x.L(B, x) \\ \text{Max} : & \quad \forall x.L(x, T) \\ \text{Tot} : & \quad \forall x.\forall y.(L(x, y) \vee L(y, x)) \\ \text{Tra} : & \quad \forall x.\forall y.\forall z. ((L(x, y) \wedge L(y, z)) \rightarrow L(x, z)) \end{aligned}$$

Let $S = \{\text{Min}, \text{Max}, \text{Tot}, \text{Tra}\}$.

1. Let $D = \{b, u, t\}$ be a set of three elements. List all possible FOL structures (D, α) of language $\mathcal{L}$ in which the set of formulas S is true and where $\alpha(B) = b$ and $\alpha(T) = t$. For each structure, show the two-dimensional table of the form:

L	b	u	t
b			
u			
t			

filled with 0 and 1 where 1 indicates the relation holds and 0 that it does not hold. Explain why these are the only structures with the required property. For each structure, indicate whether L satisfies the axioms of a partial order (reflexivity, antisymmetry, transitivity).

2. Now consider an extended language $\mathcal{L}' = \mathcal{L} \cup \{E\}$ where E is a binary relation symbol. We will define a new structure α' that also defines E . In each of the structures identified in Q1, give a table for E such that the following added formula is true:

$$\text{Both} : \forall x. \forall y. (E(x, y) \leftrightarrow (L(x, y) \wedge L(y, x)))$$

In each case, check whether the relation given by $\alpha'(E)$ is an equivalence relation (i.e., it is reflexive, symmetric, and transitive). Let $S' = S \cup \{\text{Both}\}$.

3. Does there exist a structure α' (with arbitrary domain) that makes S' true such that the relation $\alpha'(E)$ is *not* an equivalence relation?
4. (Intentionally open-ended) Can you use structure α' with domain D that satisfies S' to define a new structure α_1 that also satisfies S' , that has a possibly different domain D_1 but with size no larger than D , and where $\alpha_1(L)$ is a partial order relation?